



טופס הגשת הצעת מחקר

תוכנית למימון מחקרי הוכחת היתכנות במוסדות המתקצבים ע"י ות"ת- מחזור שני

מרכז הפרויקט: ד"ר אלכסנדר אפארצין

מוסד מגיש: מכון טכנולוגי חולון (HIT)

נושא המחקר: CareLoop: פלטפורמת בינה מלאכותית לניהול הנחיות טיפול ומעקב רפואי אמין

תחום: Artificial_Intelligence_(Applications)

Research Title: CareLoop: An AI Platform for Clinical Intent Management and Reliable Follow-Up Care

Principal Investigator: Dr. Alexander Apartsin

Institute: Holon Institute of Technology (HIT)

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A. ABSTRACTS

Hebrew Abstract (up to 1,500 characters)

מערכות רשומה רפואית אלקטרונית (EHR) מתעדות רשומות ועובדות קליניות מובנות, אך שכבה תפעולית חשובה נותרת לעיתים קרובות בטקסט החופשי בלבד: הכוונה הקלינית, כלומר, מה אמור להתבצע בהמשך עבור המטופל. הנחיות כגון "לחזור על בדיקת MRI בעוד שבועיים אם התסמינים נמשכים" עלולות לפיכך שלא להיקלט במערכות המשמשות לזימון תורים, מעקב ובקרה.

CareLoop ממירה הנחיות כאלה לפעולות מובנות, מקושרות למקור וניתנות לבקרה. החידוש המרכזי הוא בגישת ניהול כוונה קלינית (Clinical Intent Management, CIM): שכבת תשתית גנרית בין הטקסט הקליני והטרוגני לבין תהליכי העבודה הפועלים על בסיסו. הכוונה הקלינית באה לידי ביטוי בהפניות, בהוראות, בתוכניות מותנות ובהנחיות מעקב; ה-PoC מיישם אותה תחילה על הנחיות מעקב המערכת מבוססת על שיטה נירר-סימבולית לחילוף ולנרמול של פעולות מעקב ומועדיהן, שפותחה במחקרו הקודם של החוקר הראשי.

אי-הוודאות המרכזית שנותרה היא האם השיטה עוברת באופן אמין ממאגר מבוקר לטקסט קליני אמיתי והטרוגני ויכולה להשתלב בתהליך עבודה מעשי של EHR. ה-PoC ירחיב את השיטה לחילוף שדות נוספים של הכוונה הקלינית, יבצע אימות על טקסטים קליניים אמיתיים, מגוונים ומותממים, וידגים אינטגרציה מבוססת FHIR/SMART. התוצר יהיה אב-טיפוס מאומת לקראת פיילוט לניהול אמין של הנחיות מעקב.

English Abstract (up to 1,500 characters)

Electronic Health Record (EHR) systems capture records and structured clinical facts. Still, a third operational layer often remains buried in free text: clinical intent, or what is supposed to happen next for the patient. Follow-up instructions such as “repeat the MRI in two weeks if symptoms persist” may therefore never enter the systems used to schedule, track, and audit care.

CareLoop converts such instructions into structured, source-linked, and reviewable actions. We term this capability Clinical Intent Management (CIM): a reusable, auditable layer between heterogeneous clinical text and the workflows that act on it. Clinical intent spans referrals, orders, conditional plans, and follow-up; this PoC instantiates CIM on follow-up as its first concrete case.

CareLoop builds on the PI's neuro-symbolic method for extracting and normalizing follow-up actions and timing. The key remaining uncertainty is whether this approach transfers reliably from a controlled benchmark to heterogeneous real clinical text and can support a practical EHR workflow. The PoC will extend the method to additional intent fields, validate it on real de-identified clinical notes, reusing the PI's CIRCA annotation and FHIR-mapping assets, using a locked test and calibrated review, and demonstrate standards-based integration through FHIR/SMART workflows. The outcome will be a validated pre-pilot prototype suitable for design-partner evaluation and preparation of a real-world pilot.

Relevance of the Proposed Research with the Selected Topic (up to 500 characters)

Primary field: Artificial Intelligence (Application). CareLoop uses AI to turn free-text follow-up instructions into structured, trackable actions. The underlying method has already been developed and evaluated in the PI's previous research. This project validates and productizes it on real clinical text.

Secondary relevance: Medicine (patient safety, continuity of care).

Applicability Abstract (up to 1,000 characters)

Current EHR systems often leave patient-specific follow-up intent in free text, outside the structured workflows used to schedule, track, and audit care. CareLoop applies AI, combining clinical language models with deterministic normalization, to convert follow-up instructions into source-linked, reviewable actions that integrate with existing EHR workflows through FHIR, SMART on FHIR, and CDS Hooks. Initial buyers are expected to be health systems and health plans/HMOs, with quality, safety, and care coordination functions; operational users are teams responsible for closing follow-up loops. The PoC will validate real-text accuracy, review burden, interoperability, and site-level economics, while customer discovery will define pilot requirements and seek pilot commitment or LOI. Commercialization will proceed first through design-partner pilots and direct deployment, followed by licensing/OEM with EHR or workflow vendors, or an HIT spin-out if traction supports it.

Intellectual Property and Commercial Interests

Are there any approved or submitted Patents? ☐ Yes ☒ No

If yes, please provide details: [Click or tap here to enter text.](#)

Are there any industrial or commercial interests? ☐ Yes ☒ No

If yes, please provide details: [Click or tap here to enter text.](#)

Ethical and Regulatory Requirements

Does the research involve any of the following?

Please select one option in each row.

	Yes	No
Human participants	☐	☒
Animals	☐	☒
Protected natural assets	☐	☒
Plants	☐	☒
Schools	☐	☒

☒ I acknowledge that I am responsible for ensuring that all required approvals are valid and, when necessary, renewed throughout the research period.

Will the proposed research use infrastructure operated by international organizations of which Israel is a member? ☐ Yes ☒ No

If yes, please specify the infrastructure: [Click or tap here to enter text.](#)

Will the proposed research use institutional or national infrastructure? ☐ Yes ☒ No

If yes, please specify the infrastructure: [Click or tap here to enter text.](#)

B. THE RESEARCH GROUPS

1. Principal Investigator

Role in the Project: Principal Investigator (PI)

Name: Alexander Apartsin

Institute: Holon Institute of Technology

Email: alexanderap@hit.ac.il

[Click or tap here to enter text.](#)**Mobile Phone Number:** +972-54-4663885

Expertise: Artificial Intelligence, LLMs, New Product Development

2. Research Team

2.1 Role in the Project: Research Assistant/Student

Name: TBD

Institute: TBD

Email: TBD

Mobile Phone Number: TBD

Expertise: Software and model development

2.2 Role in the Project: Clinical Advisor (Senior Clinical and Clinical-Informatics Consultant)

Name: Prof. Rafael Barkan

Institute: Holon Institute of Technology

Expertise: Clinical workflow and follow-up care; clinical informatics; health-information-system integration

Role: Provides clinical governance for the PoC, validating the clinical relevance of the Actionable Clinical Record, approving annotation guidelines and adjudication of ambiguous cases, assessing clinical safety and review burden, and guiding requirements for a subsequent clinical pilot

C. RESEARCH PROPOSAL FORM - FULL PROPOSAL

The Research Plan: Up to 5 pages, including illustrations and graphs, a work plan, and a timeline (Gantt), not including the bibliography.

The research plan should include: a description of the project and the research question, existing market barriers, the innovation level of the proposed solution, its advantages over other potential solutions, a description of the expected impact of the proposed solution, ROI analysis, milestones for the progress of the project, and a description of the planned final outcome/product.

1. Research Plan

Problem, Prior Research, and POC Objectives

The problem. Clinicians constantly decide what should happen next (e.g., a repeat test, a scan, a referral), but write these as sentences, so the systems that schedule, order, remind, and audit rarely receive them in a usable form. As a result, follow-up is frequently missed, delayed, or misinterpreted, a recognized patient-safety and cost problem and a contributor to diagnostic error; ambulatory test-result follow-up and referral completion are well-documented weak points [2,5,11].

What CareLoop is and the method it uses. CareLoop is a Clinical Intent Management (CIM) platform: it makes clinical intent, what should happen next for the patient, computable. While clinical intent appears across many use cases [16], this POC starts with a focused but high-value subcase: extracting clinical follow-up instructions. For a follow-up instruction such as "repeat the blood test in two weeks," it produces a structured, source-linked Actionable Clinical Record (ACR) that includes the action, a normalized due date when the text provides one, the source span it came from, and a confidence flag for review. The method behind this was built in our previous research [1]: a hybrid design in which a learned component finds and links the action and its time expression, an explicit symbolic component normalizes the action and converts relative timing ("in two weeks" → a calendar date), and a selective-prediction step routes low-confidence cases to a clinician. CareLoop is an applied AI system: language models fine-tuned on task-specific labeled data perform extraction; an explicit symbolic layer performs normalization; and general-purpose LLMs support data augmentation and annotation pre-labeling rather than the output path.

Existing market barriers. The initial economic buyer is expected to be hospital or HMO quality, safety, and care-coordination units, while operational users are clinicians and staff responsible for completing follow-up. The unresolved upstream problem is converting follow-up intent already present in clinical text into a reliable workflow object without adding clinician data-entry burden. Adoption depends on three factors: reliable extraction from heterogeneous clinical text, integration with existing EHR workflows, and sufficient precision and provenance to support clinical trust. The PoC directly tests these three barriers.

The POC question is whether the fixed hybrid extraction and normalization approach from [1], extended using established supervised methods, can produce source-supported follow-up records from real de-identified clinical notes at a level of precision and review burden suitable for a pre-pilot system, and whether accepted records can be carried without loss of required information into a standards-based clinical workflow.

Innovation and what the POC extends. The proposed innovation is a system that converts free-text clinical intent into a structured, source-linked Actionable Clinical Record, using hybrid AI extraction, explicit normalization of actions and timing, and standard interfaces for integration into clinical workflows. The design is deliberately hybrid rather than purely generative: large language models provide breadth and language coverage, while an explicit symbolic layer ensures the normalized action, computed date, and source span are exact, so every ACR is auditable rather than a fluent guess. The innovation therefore lies not in task tracking itself, but in creating a reusable, auditable Clinical Intent Management layer between heterogeneous clinical text and downstream care workflows. This positions CareLoop within a broader shift from systems that record what happened, to systems that represent what should happen next [16]. On the controlled 2,000-note benchmark in [1], the completed method achieved action–time Pair F1 of 0.997 on seen and 0.986 on unseen

action types, outperforming the evaluated LLM baselines (Pair F1 0.51–0.57). These results establish the technical starting point; the present POC tests whether the approach transfers to heterogeneous real clinical text. It extends the same established approach to an enriched ACR comprising target, actor, condition, and dependency fields, to heterogeneous real clinical text, and to calibrated review and standards-based integration.

Advantages over existing approaches. Follow-up today depends on a clinician manually entering a task or order. Workflow and follow-up tracking systems monitor such tasks well, yet they presuppose that a structured task already exists. Rule-based and NER clinical NLP extracts entities and relations, auditable but stops short of an executable action bound to its timing. General-purpose LLM summarization is flexible, yet its output is neither reliably structured nor traceable to the source; on the PI's CIRCA benchmark, strong LLMs given the gold span recovered all required intent fields only 18–35% of the time [23]. Commercial radiology systems such as PowerScribe One do extract follow-up recommendations into structured fields and pass them downstream [21] but operate inside the radiology reporting workflow. CareLoop targets the gap these leave by recovering intent from clinical text already written across specialties and delivering it as a source-linked record that carries action, timing, conditions, and dependencies, with calibrated review and deterministic mapping into standard workflow resources.

Expected impact. CareLoop is intended to make follow-up instructions currently embedded in text available within existing tracking, reminder, and audit workflows. Prior structured follow-up programs have produced substantial improvements in completion of recommended imaging; the FIND program, for example, increased completion from 30.8% to 50.7% [12]. CareLoop addresses the upstream information gap required to supply those workflows directly from free clinical text; downstream clinical and operational effects would be evaluated in a subsequent real-world pilot.

ROI. Economic feasibility will be assessed at the site level using follow-up volume, baseline non-completion, expected improvement, value per completed follow-up, and software plus human-review costs. The primary output will be a break-even threshold, expressed as the number or percentage of additional completed follow-ups required for a site to cover CareLoop operating costs. Task 6 will estimate these inputs using customer discovery and prospective partner data, with published results such as [12] serving only as external benchmarks. Conservative, base, and higher-impact scenarios will be compared together with per-site, per-volume, and licensing revenue models.

Planned final product. By month 24, the POC will deliver a validated pre-pilot CareLoop prototype with validated extraction of the core follow-up record and evaluated extraction of target, actor, condition, and dependency fields, standards-based EHR integration, and a secure hosted review workflow, along with the market, IP, and ROI evidence needed for a clinical pilot and commercialization. This POC establishes the first focused implementation of the broader Clinical Intent Management vision, which can later extend to additional clinical workflows and intent types.

IP and commercialization plan. A freedom-to-operate review covers the POC assets, and where it supports one, a provisional patent is filed via a patent attorney. The closing output is a go-to-market and licensing plan that names the initial segment and route, along with an asset register documenting ownership and licensing status for the ACR schema, annotation guidelines, trained and calibrated models, integration components, and

project-specific derived annotation assets.

External evidence of market need. Follow-up reliability is already recognized as an operational quality problem requiring dedicated workflows. The American College of Radiology reports that up to 10% of radiology reports contain follow-up recommendations and that approximately half of the recommended examinations are not completed; it operates a national Recommendations Follow-Up Improvement Collaborative addressing this gap [4]. CMS separately treats Closing the Referral Loop as a high-priority quality measure [20]. Recent clinical informatics literature describes systematic follow-up management as a workflow spanning identification, communication, tracking through to completion, and outcome assessment. It identifies AI as a means to reduce manual effort across these stages [19]. These signals establish an existing institutional need; Task 6 therefore focuses on validating CareLoop-specific hypotheses regarding buyers, workflows, integrations, pricing, and pilot.

Project Goals and Tasks

The plan is organized as six tasks in a dependency chain: the gold dataset (Task 1) enables engineering adaptation (Task 2) and validation (Task 3), which enable EHR integration (Task 4) and hosted deployment (Task 5); market validation and commercialization (Task 6) run alongside. Validation targets are fixed before any model adaptation. Tasks 1–5 are carried out using publicly available and/or credentialed de-identified research datasets, providing a complete technical validation path. Task 6 complements this work through partner-based market discovery and product feedback, helping assess workflow fit, pilot readiness, and commercialization potential, without involving patient data.

Task (months)	Objective	Scope	Milestone
T1: Data preparation & labeling	Create a gold-standard ACR dataset on real de-identified text	Harmonize CLIP [13], i2b2 2012 [15], and MIMIC-IV [14]; define guidelines; double-label and adjudicate.	M8 frozen annotation guideline and gold dataset
T2: Engineering adaptation & extended ACR	Extend from action-time to an enriched follow-up ACR on real text	Extend the ACR with target, actor, condition, and dependency, plus provenance and confidence.	M12 extended-ACR prototype
T3: Model validation	Demonstrate reliable, calibrated performance on real text	Locked single-use test, baselines and ablations, calibration and selective review, clinician adjudication.	M16 locked validation completed against the success criteria
T4: Integration	Enable standards-based EHR integration.	Map ACRs to FHIR ServiceRequest/Task with Provenance and a SMART on FHIR review surface; test conformance.	M19 minimum FHIR integration with round-trip conformance
T5: Hosted prototype deployment	Demonstrate a triable pre-pilot prototype.	Secure, access-controlled hosted prototype on de-identified data; simulated workflow, no autonomous write-back.	M21 hosted pre-pilot prototype, externally evaluable
T6: Market validation	Validate demand, defensibility, and route to market	Customer discovery, design-partner and pilot requirements, LOI, FTO and commercialization strategy.	M24 market validation, ROI, IP/FTO, and pilot-preparation package

Tasks in Detail

Task 1. Gold-standard data. No public corpus contains the full follow-up ACR labels, so the measurement instrument is a human-labeled gold standard built on de-identified CLIP [13], MIMIC-IV-Note [14], and i2b2/n2c2 2012 [15] (reusing the CLIP action taxonomy and i2b2/THYME TIMEX3 guidelines to reduce cost). Annotation reuses the pipeline from the PI's CIRCA benchmark [23], which harmonized five clinical corpora into 10,011 FHIR-aligned intent records via three-model consensus with human adjudication, so the labeling workflow, crosswalks, and cost model are already demonstrated at scale. A pilot prevalence estimate sets the sampling allocation; if eligible instructions prove sparse, development data are enriched from higher-prevalence components and may be augmented with LLM-generated synthetic communications [22], while a separately sampled natural-prevalence test set of real text is retained for unbiased evaluation. Trained clinical-background annotators mark all eligible instructions on sampled source notes (negatives and non-actionable mentions retained, so detection recall and false-positive rate are measurable) and annotate the semantic fields; the guideline is frozen after a pilot. A stratified subset is independently double-annotated and agreement computed before adjudication, with the clinical advisor resolving disagreements. The target corpus comprises approximately 1,500–2,000 notes and at least 1,000 eligible instructions, including negatives. Target agreement span $F1 \geq 0.85$, $\kappa/\alpha \geq 0.75$, and ≥ 0.90 on normalized dates; falling short triggers guideline revision and a second pilot.

Task 2. Engineering adaptation and extended-ACR implementation. Adaptation uses supervised transformer-based clinical language models, with large language models used for candidate generation, annotation pre-labeling, and data augmentation. At the same time, the calibrated hybrid model remains the final extractor so outputs stay auditable. The action–time engine is extended to the remaining ACR fields and adapted to real text using established extraction and normalization techniques; the scientific architecture, learning objective, and temporal-normalization method are fixed, and the PoC addresses implementation, adaptation, and validation rather than development of a new basic method:

Each added element uses an established technique: supervised relation extraction for actor and dependency; span classification and typed NER for condition and target; ontology mapping to SNOMED CT and LOINC for action normalization; TIMEX3 normalization for timing; and selective prediction with probability calibration for confidence and review flags.

Task 3. Locked validation. The corpus splits into train/adaptation, validation, and a single-use test; patients/notes are disjoint. The guidelines, unseen-type vocabulary, configuration, baseline prompts/rules, and confidence threshold are frozen before the test, which is never used for tuning. Experiments cover zero-adaptation transfer, then adaptation, then one locked test; unseen types and source-supported fields; ablations; rule- and fixed-version LLM baselines on identical notes; threshold selection and risk coverage; and clinician workflow with FHIR conformance. On the locked patient-/note-disjoint test set, CareLoop must demonstrate both reliable detection and a practical level of human review. Eligible follow-up instructions must be detected with recall ≥ 0.80 . Among detected records, the selective-prediction system must achieve precision ≥ 0.95 while automatically accepting at least 70% of records without manual review. Temporal normalization must match the gold-standard date, interval, or event-relative representation under the predefined normalization policy for

≥90% of resolvable expressions; absolute date error will additionally be reported for expressions resolving to a specific calendar date. We will report detection precision/recall/F1, selective coverage, calibration, per-field performance, and performance on predeclared unseen action types with 95% confidence intervals. The integration gate requires that mandatory information be preserved in ACR→FHIR→ACR round-trip tests with no critical FHIR conformance failures.

Task 4. Integration. Reviewed ACRs will be mapped deterministically to a minimum FHIR workflow representation consisting of ServiceRequest/Task with Provenance, preserving source linkage and confidence. A SMART on FHIR application will provide the clinician-facing review surface. Conformance and ACR→FHIR→ACR preservation will be tested using standard FHIR tooling. CDS Hooks, CarePlan, CommunicationRequest, and additional workflow resources will be added only where required by the design-partner pilot specification.

Task 5. Hosted prototype deployment. The prototype is deployed as a containerized, access-controlled hosted service supporting note and FHIR ingestion, model inference, clinician review with source-linked accept/edit/reject controls, provenance display, and standards-based FHIR export. Deployment uses reproducible containers and audit logging on de-identified data; production-scale multi-tenancy and enterprise hardening are outside the PoC scope.

Task 6. Market validation. External evidence establishes follow-up reliability as an existing healthcare workflow and quality problem; Task 6 tests the commercial hypotheses specific to CareLoop. Structured customer discovery with health systems, health plans, care-coordination units, and EHR/workflow vendors will identify the economic buyer, operational user, current workflow and alternatives, integration requirements, acceptable review burden, procurement constraints, pricing basis, and pilot success criteria. We will seek to identify a suitable design partner and obtain an LOI, a pilot commitment, or a jointly agreed pilot specification that defines workflow, integration requirements, and success criteria. Clinical experts will participate as paid consultants evaluating de-identified artifacts; no personal research data will be collected.

2.Schedules and Work Plan (Gantt)

#	Task	1-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24
1	Data preparation & labeling								
2	Engineering adaptation (extended ACR)								
3	Model validation								
4	EHR integration (FHIR / SMART on FHIR)								
5	Hosted prototype deployment								
6	Market validation & commercialization								

*You can set different deadlines

3.Bibliography: List of articles most relevant to the research proposal. Mark with an asterisk the five sources most relevant to the proposal, excluding those authored by the applicants.

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13. (*) Mullenbach, J., Pruksachatkun, Y., Adler, S., et al. (2021). CLIP: a dataset for extracting action items for physicians from hospital discharge notes. *Proceedings of ACL-IJCNLP*, 1365–1378. Dataset version 1.0.0, PhysioNet, <https://doi.org/10.13026/kw00-z903>.

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D. BIOGRAPHICAL SUMMARY- BIOSKETCH

Name:

Dr. Alexander Apartsin

Position Title:

Senior Lecturer, School of Computer Science, Holon Institute of Technology

Education/Training:

Institution and Location	Degree	MM/YY	Field of Study
Technion, Israel Institute of Technology, Haifa	B.Sc.	1995	Computer Science
Weizmann Institute of Science, Rehovot	M.Sc.	2005	Computer Science (Computer Vision)
IEEE Computer Society	CSDP	2005	Software Development Methodologies
Polytechnic Institute of New York University	M.Sc.	2006	Technology Management and Innovation
PDMA: Product Development and Management Association	NPDP	2006	New Product Development Methodologies
Tel Aviv University, Tel Aviv	Ph.D.	2012	Computer Science, Machine Learning and Signal Processing
Weizmann Institute of Science, Rehovot	Postdoc	2013	Computer Vision and Machine Learning

A. Personal Statement

This PoC requires a combination of scientific, engineering, product-development, and commercialization capabilities. My career has developed across these areas, combining research in reliable language-information extraction with the development of operational AI systems, end-to-end new-product development, and entrepreneurship.

My academic background is in machine learning across a range of application domains. I hold a Ph.D. in machine learning and signal processing from Tel Aviv University. My doctoral research produced seven first-author publications co-authored with Nobel laureate Leon N. Cooper and was followed by postdoctoral research at the Weizmann Institute. My subsequent academic and industry work has focused on translating AI methods into reliable operational systems, including language-processing and information-extraction applications. At HIT, I lead the Pragmatic AI research program, focused on developing AI systems that remain reliable, interpretable, and usable under real-world conditions. The program combines academic research, student participation, and industry collaboration.

A substantial part of my current and previous work, in both academia and industry, has focused on reliably extracting and processing information from text using natural language processing, machine learning, and, more recently, LLMs. Further details are available at apartsin.com/research/robust-language-ai.html. In particular, my recent work provides the scientific foundation for CareLoop, including the hybrid neural-symbolic method on which this proposal builds [1].

Much of my industry career has focused on applying AI methods toward operational systems and products. At the Motorola Solutions Israel Innovation Center, I led work on AI for public-safety applications spanning image, text, video, and audio. As VP of AI at harmon.ie, I worked on large-scale enterprise document-intelligence systems to extract and organize information from unstructured document collections. At Questar, I led research and engineering teams that developed automotive telematics AI.

I have also co-founded and provided technical leadership to several technology ventures. At SearchNG, I helped establish the company and led its capital-raising activities. This practical experience is complemented by formal training in innovation and product development: an M.S. in Management with a focus on innovation from NYU Polytechnic and the PDMA New Product Development Professional (NPDP) certification, covering new-product development and commercialization from opportunity identification through launch.

Together, these experiences provide a practical basis for carrying the PoC from scientific validation through engineering implementation to a deployable product and a commercialization path.

B. Selected Positions (selected)

- 2024 to present: Senior Lecturer, School of Computer Science (HIT).
- 2020 to 2024: Head of Data Science, Questar Automotive
- 2019-2020: VP Artificial Intelligence, Harmon.ie
- 2015 to 2019: Head of AI Research, Motorola Solutions Israel Innovation Center.
- 2013-2019: AVP Data Science, Citigroup TLV Innovation Lab
- 1999-2015: Misc. software companies, roles including: Architect, R&D Manager, CTO, and Founder

C. Selected Publications Relevant to the Proposed Project

1. Laufer, M., Aperstein, Y., & Apartsin, A. (2026). Reliable Extraction of Clinical Follow-Up Instructions: A Hybrid Neural-Symbolic Pipeline. arXiv:2605.26560, <https://doi.org/10.48550/arXiv.2605.26560>. (*Not yet peer-reviewed; submitted for peer review; scientific basis for the proposal.*)
2. Atia, T., Aperstein, Y., & Apartsin, A. (2026). SeaAlert: Robust Severity Classification and LLM-Based Information Extraction for Noisy Maritime Distress Communications. IEEE Access, 14, 101292–101303. <https://doi.org/10.1109/ACCESS.2026.3709004>.
3. Aperstein, Y., Moran, E., & Apartsin, A. (2026). IRC-Bench: Recognizing Entities from Contextual Cues in First-Person Reminiscences. Machine Learning and Knowledge Extraction, 8(7), 186. <https://doi.org/10.3390/make8070186>.
4. Werthaim, M., Kimhi, M., Apartsin, A., & Aperstein, Y. (2026). A Benchmark for Evaluating Diagnostic Questioning Efficiency of LLMs in Patient Conversations. Scientific Reports, 16, 6121. <https://doi.org/10.1038/s41598-026-37022-y>
5. Aperstein, Y., Gottlib, A., Benita, G., & Apartsin, A. (2025). Explainable Semantic Text Relations: A Question-Answering Framework for Comparing Document Content. Information, 16(12), 1090. <https://doi.org/10.3390/info16121090>.
6. Apartsin, A., & Aperstein, Y. (2026). Clinical Intent Extraction: A FHIR-Aligned Representation and the CIRCA Benchmark. arXiv:XXXX.XXXXX, <https://doi.org/10.48550/arXiv.XXXX.XXXXX>. (Preprint; not yet peer-reviewed. Harmonized intent dataset and FHIR mapper underpinning Task 1.)

D. Research Support: None

E. REQUESTED BUDGET

Total funding requested: 350000 NIS

Duration (months): 24

Currency: ILS(NIS)

Category	Sum	Budget Justification
Personnel- Research Assistants, Students	150,000	Research assistant / MSc student (24 months, part-time): data preparation and labeling support, supervised model adaptation, core follow-up extraction and ACR enrichment, locked validation, FHIR integration, and prototype engineering
Technical Support and Consulting	30,000	Clinical and clinical-informatics consultation: clinical use-case definition; review of the ACR schema and annotation guidelines; adjudication of ambiguous cases; assessment of clinical validity, error categories, and review burden; workflow integration; and definition of pilot requirements and success criteria.
Professional Services	75,000	Clinical data labeling and expert annotation: 50,000. Primary evaluation uses publicly available and/or credentialed de-identified research datasets (CLIP, i2b2 2012, and MIMIC-IV-Note). Intellectual property and commercialization: 25,000, covering prior-art and freedom-to-operate review, assessment of protectable project assets, and, if justified by that assessment, drafting and filing of an appropriate patent application through a patent attorney, together with commercialization and licensing planning.
Equipment	0	Not requested
Computer Equipment	32,500	Cloud computing services (recorded under the form's Computer Equipment category): secure cloud compute for model adaptation, locked evaluation, hosted prototype deployment, and reproducible integration testing.
Consumables	0	Not requested
Administrative	0	Not Requested
Overhead (up to 15%)	52,500	Retained by HIT for support, monitoring, and oversight per the call.
Other	10,000	External prototype evaluation, integration demonstration, structured customer-validation activities, and commercialization deliverables required for pilot readiness.

F. APPLICABILITY APPENDIX

Description of the project's potential to be implemented (e.g., scalability), secure additional investment, etc. Up to 2 pages.

- Possible pathways for implementing the project (*such as licensing, selling the technology, establishing a company, etc.*).
- Existing or anticipated intellectual property related to the proposed project.
- Description of the potential customers and/or investors for the project's products.
- Expected timetable for implementing the project.
- Evaluation of the feasibility of the project.

Opportunity and Scalability

External evidence supports the market need. Follow-up reliability is already an institutional quality priority, evidenced by the ACR Recommendations Follow-Up Improvement Collaborative [4], the CMS closing-the-referral-loop measure [20], and recent work on AI-supported follow-up tracking [19]. The PoC therefore tests the remaining product-specific hypotheses on adoption, integration, pricing, and pilots.

CareLoop is a Clinical Intent Management (CIM) platform that addresses a missing layer in healthcare IT: clinical intent, meaning what is supposed to happen next for the patient. We have characterized this as a third generation of healthcare IT, moving from systems that record what happened to systems that represent what should happen next [16]. Current EHRs store records and structured facts, but follow-up instructions such as “repeat the blood test in two weeks” or “refer to cardiology if symptoms persist” often remain only in free text and therefore cannot be reliably monitored. CareLoop converts these instructions into structured, auditable, source-linked actions.

The initial product focuses on one well-defined, high-value use case: clinical follow-up instructions. It integrates with existing systems through standards such as FHIR, SMART on FHIR, and CDS Hooks, without replacing the EHR or adding new data-entry burden. The same CIM platform can later support referrals, discharge instructions, radiology recommendations, handovers, and patient messages. This creates a clear path from a focused initial product to a broader platform: start with one high-value workflow, then expand within the same customer and through EHR and workflow-vendor distribution.

The PoC is designed to validate the core CIM technology in a defined use case and, if successful, provide the basis for expansion into additional clinical workflows and a broader Clinical Intent Management platform.

Commercialization Pathways. CareLoop will follow a staged commercialization route. During the PoC, the first objective is to identify a health-system or health-plan design partner, validate workflow fit, and agree on the requirements and criteria for a real-world pilot. After a successful PoC, the first pilot can be initiated with that partner. Successful pilot evidence can then support direct deployment and creation of a reference customer. Scaling would then focus on licensing or OEM integration with EHR and clinical-workflow vendors, allowing the same CIM capability to be distributed across multiple healthcare organizations. Depending on pilot traction and partner interest, commercialization may proceed through HIT licensing or formation of a spin-out. The expected business model is recurring revenue based on site, clinical volume, or enterprise licensing.

Intellectual Property. The scientific foundation comes from our previous research: the hybrid extraction method [1] and the CIRCA harmonized intent dataset and FHIR mapper [23], both PI-owned assets the product will build on. The PoC will create product-specific assets including the ACR representation and schema, annotation guidelines and project-specific annotation layers, trained and calibrated models, evaluation infrastructure, workflow architecture, implementation know-how, and integration components.

A freedom-to-operate and IP review will determine which developments may justify patent protection and which are better retained as proprietary know-how. Defensibility is expected to arise from the combination of protectable methods where applicable, project-specific annotation and calibration assets,

validated integration architecture, and implementation know-how. Commercial rights in derived assets will remain subject to the licenses and data-use agreements governing the underlying clinical corpora.

Customers and Follow-on Investment. Initial customers are expected to be health-system quality and safety units, health plans/HMOs, care-coordination functions, and outpatient clinics. These organizations already have follow-up reliability and workflow-performance problems. EHR and care-coordination vendors represent a second customer class that can embed the CIM engine into their products.

The PoC is designed to create the evidence required for the next investment decision: validated performance on real de-identified text, clinician-reviewed outputs, a hosted prototype, standards-based integration, customer-discovery evidence, a prospective pilot or Letter of Intent, and an IP/FTO assessment.

Successful completion would move CareLoop from a research-backed concept to a validated pre-pilot health-tech asset suitable for preparing and initiating a real-world clinical pilot, supporting follow-on funding from digital-health investors, strategic healthcare partners, or EHR and healthcare-software vendors.

Expected Timetable. During PoC, CareLoop validates technical performance, clinician-facing outputs, interoperability, hosted deployment, site-level economics, and IP potential. At the same time, customer discovery identifies design partners and seeks an LOI or agreed pilot specification. In the first 12 months after the PoC, the first real-world pilot is prepared and initiated with that partner; in months 12–24 it is completed, demonstrating operational value and willingness to pay and converting validation into a reference customer. Thereafter, scaling can proceed through enterprise sales, strategic partnerships, and OEM licensing, with expansion into additional clinical-intent workflows.

Feasibility and Risk Reduction. Feasibility is supported by the results of our previous research [1], which demonstrated strong performance relative to conventional LLM baselines. The PoC therefore focuses on transfer, generalization, and productization rather than proving a new scientific principle. CareLoop’s modular architecture strengthens technical feasibility by integrating with existing clinical systems rather than replacing them.

Key risks and mitigation. The principal technical risk is lower accuracy or excessive review burden on heterogeneous real text; if triggered, the initial product will restrict supported instruction classes and route a larger fraction of uncertain records to review. Sparse condition or dependency examples will be treated as enrichment fields rather than core release gates. Integration risk is limited by the minimum FHIR workflow profile, with additional resources added only when required by a pilot. Commercial risk is addressed through customer discovery from project start, explicit testing of buyer and pricing hypotheses, and a target pilot commitment, LOI, or agreed pilot specification before project completion.